

Solving Multi-Step Equations

Equation Flip- Book Instructions:

1. Choose three pages of construction paper.
2. Fold papers hamburger in accordion style so that about 1 inch is left as a tab.
3. Title the cover "Solving Equations: An How-to Flip Book". Write your name.
4. Label tabs in this order from front to back.

Two-Step Equations

Equations with Like Terms

Equations with Grouping Symbols

Equations with Division

Equations with Variables on Both Sides

Solving Two-Step Equations Review: *Copy these examples in your flip-book.*

Example:

$$2n + 5 = 13$$

Subtract _____ from both sides.

$$2n = 8$$

Divide both sides by _____.

$$n = \underline{\hspace{1cm}}$$

Example:

$$\frac{x}{-8} - 6 = 8$$

_____ 6 to both sides.

$$\frac{x}{-8} = 14$$

_____ both sides by -8.

$$x = -112$$

Name _____

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Equations with Like Terms

Steps:

1. Combine any like terms on each side of the equal sign.
2. Undo adding or subtracting.
3. Undo multiplying and dividing.

Examples:

1. $5a - 7 + 3a = 17$

Combine like terms: $5a + 3a = \underline{\hspace{2cm}}$

Add $\underline{\hspace{2cm}}$ to both sides of the equation.

Divide both sides of the equation by $\underline{\hspace{2cm}}$.

2. $9b + 6 - 5b = 18$

3. $2y - 7 + y = 8$

4. $-5c - 10 - 5c - 2 = 8$

5. $-31x - 20 + 3x + 2 = 38$

6. $15n + 16n + 4n = 130$

7. $h - 48 + 17h =$

8. $-6g + 7g - 7 = 9$

9. $16 - 12a - 13a = -9$

10. $8y + 43 - 7y = -15$

11. $15x - 8 - 3x = 16$

12. $5n + 3 + 4n = 30$

13. $h - 8 + 7h + 2 = 42$

14. $-3g + 1 + 2g + 5 = 15$

15. $-2b + 10 - 3b - 3 = 2$

Name _____

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Equations with Grouping Symbols

Steps:

1. Use the distributive property to undo parenthesis.
2. Combine any like terms on each side of the equal sign.
3. Undo adding or subtracting.
4. Undo multiplying and dividing.

Examples:

1. $5 + 3(n + 12) + 4n = 41$

Use the distributive property: $3 \cdot n + 3 \cdot \underline{\hspace{1cm}}$

Combine like terms: $3n + 4n = \underline{\hspace{1cm}}$ and $5 + 36 = \underline{\hspace{1cm}}$

Subtract $\underline{\hspace{1cm}}$ to both sides of the equation.

Divide both sides of the equation by $\underline{\hspace{1cm}}$.

2. $9b + 6 - 5b = 18$

3. $2y - 7 + y = 8$

4. $-5c - 10 - 5c - 2 = 8$

5. $-31x - 20 + 3x + 2 = 38$

6. $15n + 16n + 4n = 130$

7. $h - 48 + 17h =$

8. $-6g + 7g - 7 = 9$

9. $16 - 12a - 13a = -9$

10. $8y + 43 - 7y = -15$

11. $15x - 8 - 3x = 16$

12. $5n + 3 + 4n = 30$

13. $h - 8 + 7h + 2 = 42$

14. $-3g + 1 + 2g + 5 = 15$

15. $-2b + 10 - 3b - 3 = 2$

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Equations with Division

Steps: (Write in FlipBook)

1. Undo dividing first when the numerator is an expression by multiplying.
2. Combine any like terms on each side of the equal sign.
3. Undo adding or subtracting.
4. Undo multiplying and dividing.

Examples:

1. $\frac{5.6-2a+1.4}{1.2} = 4.5$

Multiply both sides by _____.

Combine like terms: $5.6 + 1.4 =$ _____

Subtract _____ to both sides of the equation.

Divide both sides of the equation by _____.

Copy the example above in FlipBook

2. $\frac{6m-3}{7} = 9$

3. $\frac{\frac{1}{3}x-2}{5} = -2$

4. $\frac{17x-24}{18} = \frac{5}{9}$

5. $\frac{-2y+12-4y}{6} = -2$

$$6 \quad \frac{0.4n-1}{7} = 5$$

$$7. \quad \frac{18-4t}{0.5} = 20$$

$$8. \quad \frac{2n+3}{-12} = \frac{1}{4}$$

$$9. \quad \frac{1.4n-4.5}{1.5} = 8.2$$

Name _____

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Equations with Variables on Both Sides

Steps: (Write in FlipBook)

1. Combine any like terms on each side of the equal sign.
2. If like variables are on opposite sides, identify the like term with the lesser coefficient.
3. *Move* that term to the other side by UNDOING.
4. Undo adding or subtracting.
5. Undo multiplying and dividing.

Examples: (FlipBook)

1. $18x - 2 - 12x = -6x - 26$

Combine like terms: $18x - 12x =$ _____

The like term with the lesser coefficient is _____.

Move _____ by adding it to both sides.

Add _____ to both sides.

_____ both sides by _____.

2. $15t = 17 + 8t - 24$

3. $\frac{8}{15}x = 8 - \frac{2}{15}x$

4. $6k - k = 27 - 4k$

5. $-3w + 6 = 8w + 9 - 2w$

6. $\frac{1}{4}n + 10 = \frac{2}{3}n$

7. $10 + 8c + 15 = -2c - 40$

8. $0.8a - 6 = -0.6a + 22$

9. $6a - 42 = 2a + a$

10. $-6x - 32 = 2x$

11. $32 - 5x = 3x + 8$